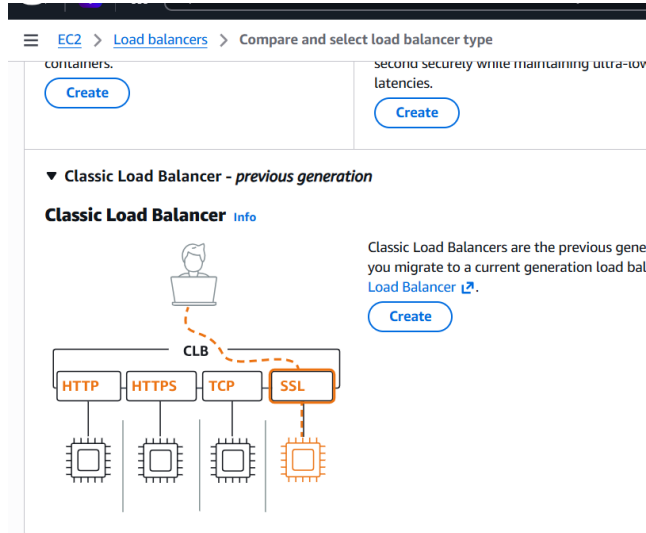


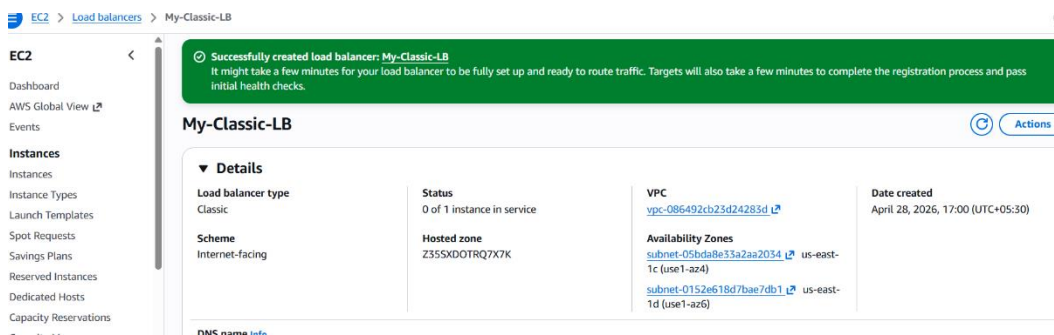
TASK ON LOAD BALANCER

1. Configure Classic Load balancer.

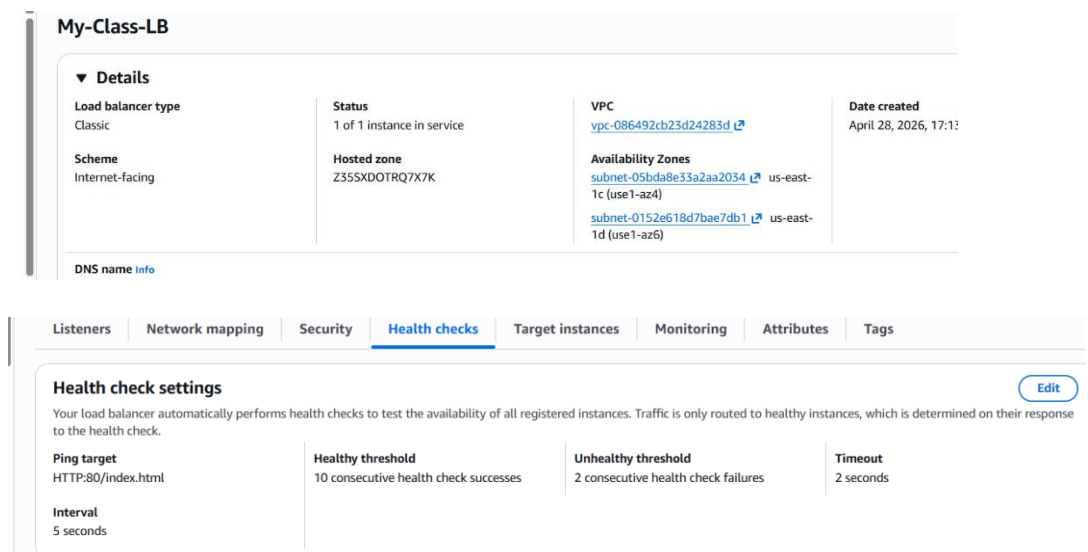
- Created Classic Load balancer from dashboard load balancer



- Created a new classic load balancer by clicking above create



- Later attached instances to it



2. Configure Application Load balancer.

- Created Application load balancer from dashboard

My-AppLoadBalancer

▼ Details			
Load balancer type Application	Status Active	VPC vpc-086492cb23d24283d	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-05bda8e33a2aa2034 us-east-1c (use1-az4) subnet-0152e618d7bae7db1 us-east-1d (use1-az6)	Date created April 27, 2026, 16:40 (UTC+05:30)
Load balancer ARN	DNS name Info		

- Attached instances and also target group to it

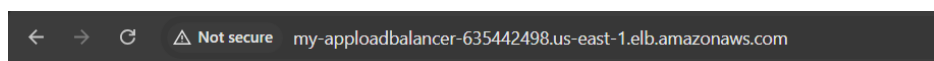
Listeners and rules (1) Info Manage rules

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the rules.

Filter listeners

☐	Protocol:Port	Default action	Rules	ARN
☐	HTTP:80	Forward to target group My-TGW: 1 (100%) Target group stickiness: Off	1 rule	ARN

- Checked the url of app load balancer



Public EC2 Web Server

This is running from Apache on AWS!

3. Configure Network Load balancer.

- Created a Network Load balancer

EC2 > Load balancers > My-Network-LB

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types

✓ **Successfully created load balancer: My-Network-LB**

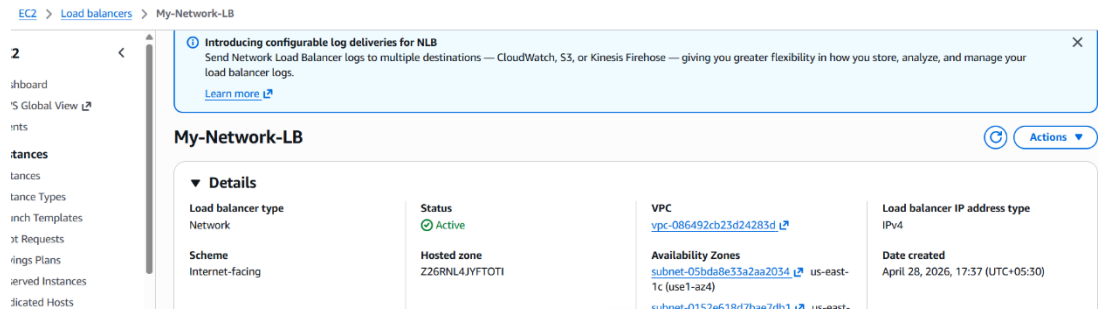
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial checks.

🔔 **Introducing configurable log deliveries for NLB**

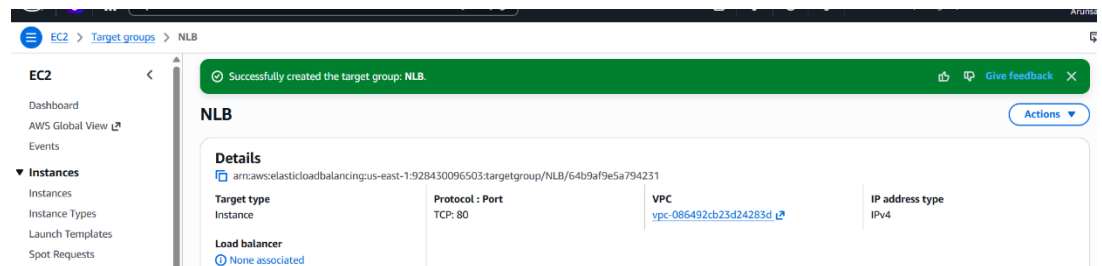
Send Network Load Balancer logs to multiple destinations — CloudWatch, S3, or Kinesis Firehose — giving you greater flexibility in how you store, analyze, and manage your load balancer logs.

[Learn more](#)

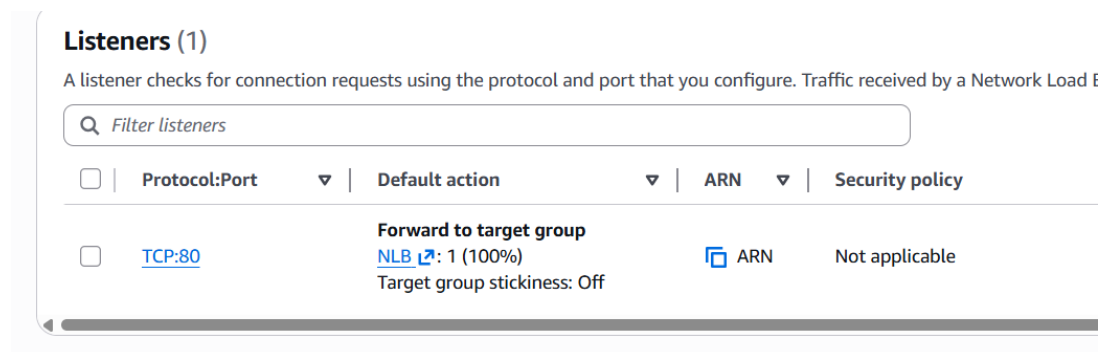
- Given instances to it



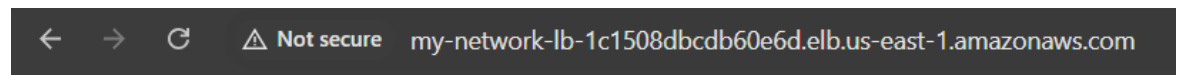
- Created a new target group for that TCP 80 port



- We can check in listeners of NBL



- Copied the NBL url and check in the browser

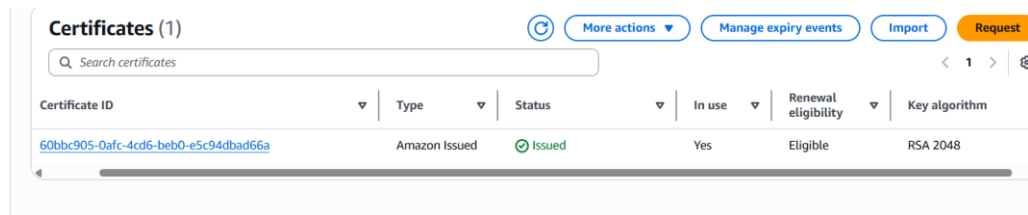


Public EC2 Web Server

This is running from Apache on AWS!

4. Attach SSL for application load balancer.

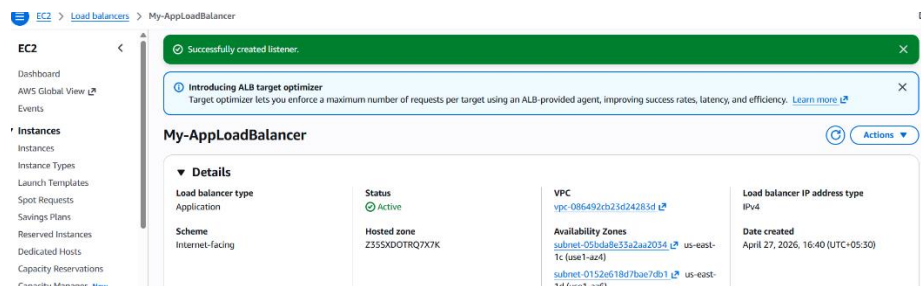
- Create ACM request to load balancer



The screenshot shows the 'Certificates (1)' page in the AWS IAM console. It features a search bar, a table with columns for Certificate ID, Type, Status, In use, Renewal eligibility, and Key algorithm, and buttons for 'More actions', 'Manage expiry events', 'Import', and 'Request'.

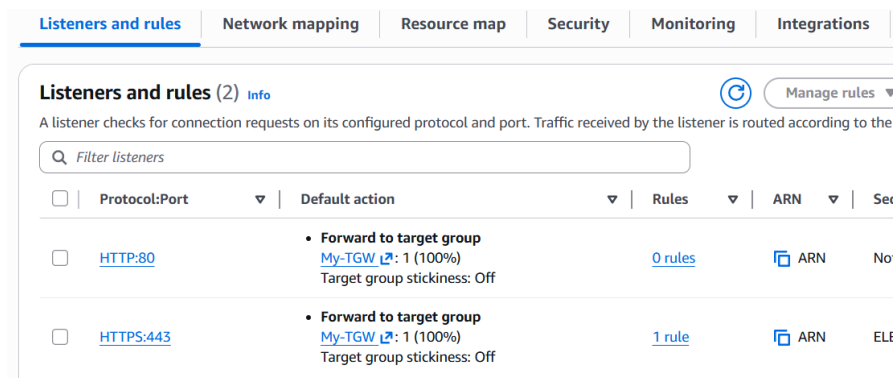
Certificate ID	Type	Status	In use	Renewal eligibility	Key algorithm
60bbc905-0afc-4cd6-beb0-e5c94dbad66a	Amazon Issued	Issued	Yes	Eligible	RSA 2048

- Added new listener to App-load-balancer HTTPS



The screenshot shows the 'My-AppLoadBalancer' details page in the AWS Management Console. It includes a 'Details' section with fields for Load balancer type, Scheme, Status, Hosted zone, VPC, Availability Zones, Load balancer IP address type, and Date created. A green notification banner at the top states 'Successfully created listener.' and a blue banner below it introduces the 'ALB target optimizer'.

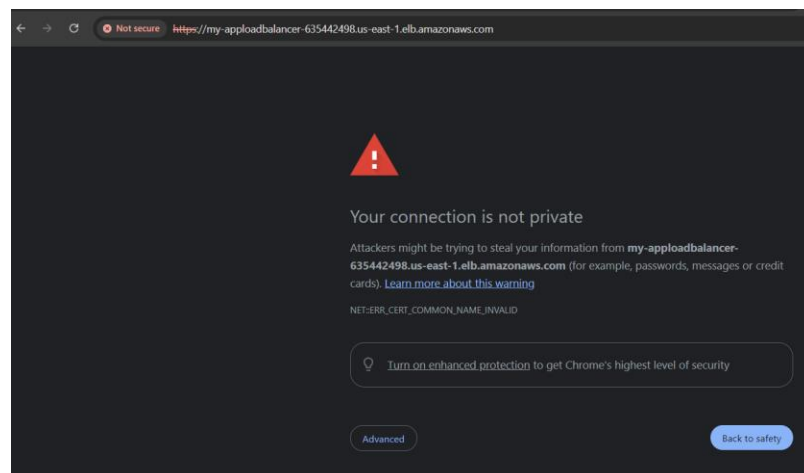
- HTTPS 443 created



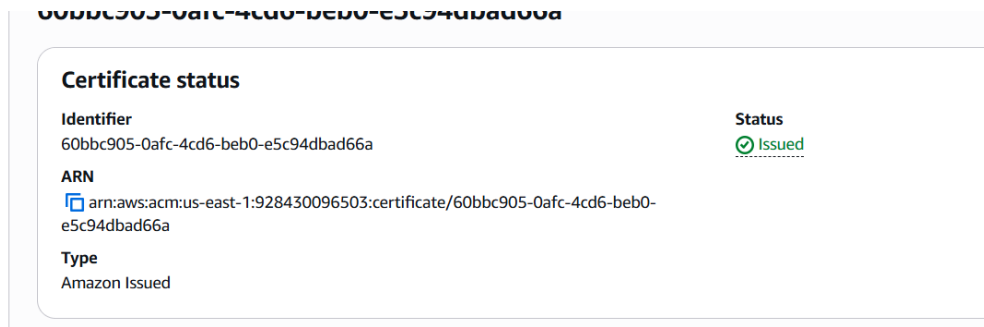
The screenshot shows the 'Listeners and rules' page for 'My-AppLoadBalancer'. It displays a table with columns for Protocol:Port, Default action, Rules, ARN, and Set. Two listeners are listed: HTTP:80 and HTTPS:443. The HTTPS:443 listener has one rule, 'Forward to target group', which is linked to 'My-TGW' with 100% weight and 'Target group stickiness: Off'.

Protocol:Port	Default action	Rules	ARN	Set
HTTP:80	Forward to target group My-TGW: 1 (100%) Target group stickiness: Off	0 rules	ARN	No
HTTPS:443	Forward to target group My-TGW: 1 (100%) Target group stickiness: Off	1 rule	ARN	ELB

- Checked My-app-load-balancer DNS we get secure error

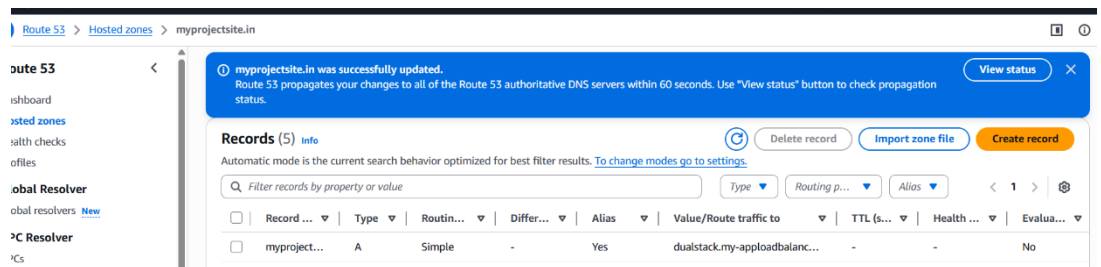


- Checked certificate url the amazon issued

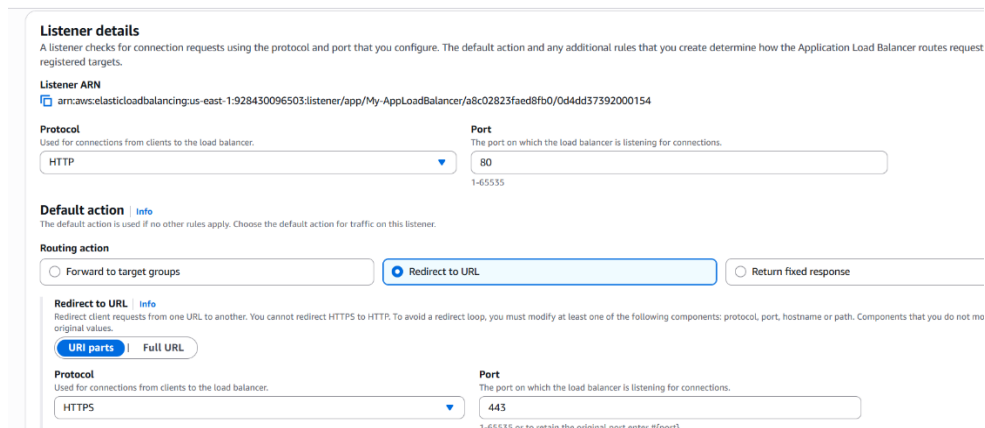


5. Map Application load balancer to R53.

- In Route 53 added new record with alias and attached app-load-balancer



- In My-app-load-balancer change the http port to redirect url and save it



- And checked my domain in the website and it worked correctly.



Public EC2 Web Server

This is running from Apache on AWS!

6. Push the application load balancer logs to S3.

- Created a new bucket in s3 with unique name for app-lb

currency			(UTC+05:30)
Access Points			
Access Points for FSx			
Access Grants			
Access Analyzer			

☐	my-alb-log-873	US East (N. Virginia) us-east-1	April 27, 2026, 17:08:36 (UTC+05:30)
☐	my-app-lb-logs-333	US East (N. Virginia) us-east-1	April 28, 2026, 21:51:37 (UTC+05:30)

- Add the policy in permissions

Amazon S3 > Buckets > my-app-lb-logs-333

Successfully edited bucket policy.

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts.

Public access is blocked because Block Public Access settings are turned on for this bucket
To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about using Amazon S3](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowALBAccessLogs",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::127311923021:root"
      },
      "Action": "s3:PutObject",
      "Resource": "arn:aws:s3:::my-app-lb-logs-333/AWSLogs/928430096503/*"
    }
  ]
}
```

- In App-load-balancer modified the attribute for new bucket of s3

Monitoring

☒ **Access logs**
Access logs deliver detailed logs of all requests made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

S3 URI
 [View](#) [Browse S3](#)
Format: s3://bucket/prefix

☐ **Connection logs**
Connection logs deliver detailed logs of all connections made to your Elastic Load Balancer. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

☐ **Health check logs - new**
Health check logs deliver detailed logs of health checks for all targets in your Elastic Load Balancer's target groups. Choose an existing S3 location. If you don't specify a prefix, the logs are stored in the root of the bucket. Additional charges apply. [Learn more](#)

- Modified and saved

EC2 > Load balancers > My-AppLoadBalancer

EC2

Dashboard
AWS Global View
Events

instances
instances
instance Types
Launch Templates

Successfully modified load balancer attributes.

Introducing ALB target optimizer
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

My-AppLoadBalancer [Actions](#)

Details

- After some we can check in s3 of ne bucket automatic logs created

≡ [Amazon S3](#) > [Buckets](#) > my-app-lb-logs-333

my-app-lb-logs-333 Info

[Objects](#) | Metadata | Properties | Permissions | Metrics | Manag

Objects (1)   Copy S3 URI (

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) 

 Find objects by prefix

☐	Name	▲	Type	▼
☐	 AWSLogs/		Folder	

- Open log file we can see ELB

≡ [Amazon S3](#) > [Buckets](#) > [my-app-lb-logs-333](#) > [AWSLogs/](#) > 928430096503/

928430096503/

[Objects](#) | Properties

Objects (2)   Copy S3 URI  Copy URL

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#)  to get a list of : [more](#) 

 Find objects by prefix

☐	Name	▲	Type	▼	Last modified
☐	 elasticloadbalancing/		Folder		-
☐	 ELBAccessLogTestFile		-		April 28, 2026